

AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

46

steering_knuckle_RH

116 × 237 × 210 mm · 356.1 cm³ · AI 0.962 kg / Steel 2.795 kg

Manufacturability: 46/100 · Confidence: High

§1 — Geometry & Part Summary

Bounding Box	116.5 × 236.8 × 210.1 mm	Surface Area	789.3 cm ²
Volume	356.10 cm ³	Weight (AI)	0.962 kg
Weight (Steel)	2.795 kg	Weight (Plastic)	0.374 kg

§2 — Detected Features

Feature Type	Count	Significance	Description
Cylindrical bores/bosses	124	High	124 cylindrical faces incl. 14 holes (r 5–7.5mm) and 3 boss/shaft features (r 12.5–39mm) — likely bearing bore, tie-rod & ball-joint bores
Threaded features	1	High	Threaded holes detected, requiring tapping/thread-milling post-cast
Undercuts	52	High	52 undercut faces requiring side-cores or re-orientation in the mould
Free-form B-spline surfaces	24	Medium	24 free-form surfaces — likely blend/fillet transitions typical of a knuckle arm casting
Planar machined pads	99	Medium	99 planar faces, many likely as-cast draft faces with a subset requiring machined datum/mating pads

§3 — Material Analysis

AI-suggested

Primary Material:

Ductile (spheroidal graphite) iron GJS500-7 (60% confidence)

Part is a steering knuckle — a suspension/drivetrain component subjected to high cyclic loads. Combined with the cast_and_machine forced process and 200,000 units/year volume, this profile matches standard automotive practice: sand or high-pressure sand-moulded ductile iron casting followed by CNC finish machining of bores and mating faces. Ductile iron offers the strength/fatigue performance needed for a knuckle at a fraction of steel forging cost, and is far more consistent with the wall-thickness/undercut profile here than aluminium (no HPDC-thin-wall or lightweighting signal present).

Alternatives:

Ductile iron GJS600-3 (higher strength) (25%) · Austempered ductile iron (ADI) (10%) · Aluminium casting alloy LM25 (5%)

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Sand casting + CNC finish machining	cast_and_machine	85%	0.2960	User-forced process; geometry (52 undercuts, non-uniform thick sections, 3 machining setups, 14 holes) is classic near-net sand-cast knuckle with post-machining of bores/pads.

§5 — Manufacturability Risks (Score: 46/100)

Severity	Feature / Area	Description	Recommended Action
High	Undercuts (52 faces)	52 undercut faces relative to a single draw direction will require side-cores/slides in the sand tooling or multiple machining re-orientations.	Consolidate undercuts onto fewer mould-parting directions or design them as post-cast machined features to simplify the pattern/core-box.
High	Zero-draft faces (102)	102 faces show zero draft, which will cause pattern sticking and core damage during sand-mould stripping, raising scrap and rework.	Add minimum 1–3° draft to all non-critical faces before pattern release.
Medium	Thin wall (min 1.36mm)	Minimum wall of 1.36mm is well below the safe fill thickness for ferrous sand casting (typically 4–6mm), risking cold-shuts, misruns or porosity.	Increase local minimum wall to "e 4 mm ribbing instead of thin unsupported webs.
Medium	Bearing/ball-joint bores (r 31.5–37.5mm)	Large bores need generous as-cast machining stock and tight post-machining tolerance/finish for bearing seating.	Cast bores 2–3mm undersize on diameter and finish-bore in a single fixtured operation to hold concentricity.
Low	Free-form B-spline surfaces (24)	Free-form blends increase pattern/core CNC-machining cost and complicate CMM inspection.	Simplify blends to standard fillet radii where not load-critical.

§6 — DFM Issues (cast_and_machine)

Severity	Area	Description	Impact	Fix
High	Undercuts / draft	52 undercut faces and 102 zero-draft faces conflict with a single mould draw direction, forcing side-cores or split patterns.	Increases pattern/tooling cost and cycle time; risk of core breakage and scrap.	Re-orient part or add draft/split-line features to remove undercuts before tooling design.
Medium	Thin wall sections	Local wall thickness down to 1.36mm is below reliable ferrous sand-casting fill thickness.	Higher risk of misruns, porosity and rejected castings.	Thicken critical thin webs to "e 4 mm or replace with ribs/gussets.
Medium	Bearing bore / ball-joint bore	Large-diameter functional bores (31.5–37.5mm) are cast near-net and need controlled machining stock and fixturing to hold concentricity.	Excess or insufficient stock causes tool wear, tolerance drift, or scrap.	Specify uniform 2–3mm machining allowance on all functional bores and finish in one clamped operation.
High	Multi-setup machining	3 required setups (+Z, +X, +Y) driven by geometry orientation of undercuts/bosses increase fixture changes and cumulative positional error.	Longer cycle time, higher labour cost, and stack-up tolerance risk between setups.	Design common datum features accessible from a single 5-axis setup to consolidate operations.
Low	Free-form blend surfaces	24 B-spline surfaces complicate pattern equipment machining and	Higher pattern-making cost and inspection time.	Standardise blends to constant-radius fillets where not

Severity	Area	Description	Impact	Fix
		inspection programming.		structurally required.

§7 — Cost Range & Suggested Inputs

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£6.50	£9.80	£14.50

Net Weight	2.546 kg	Material	mat-gjs500
Recommended Process	cast_and_machine	Cycle Time	0.2161 hr/part
Setup Time	1.000 hr	Operations	0 ops

Operations Detail

Process-Specific Parameters

Casting Subtype	sand
Die/Mould Cost	£17,000
Die Life	8,000 shots
Cavities	2
Yield	65.0%
Flash Weight	0.000 kg
Yield	0.0%
Die Cost	£0
Strokes	0
Cavities	0
Wall Thickness	0 mm
Mould Cost	£0
Mould Life	0 shots
Projected Area	0.0 cm²

§8 — AI Analysis Explanation

This is a right-hand steering knuckle — a safety-critical suspension component. The sparse fill ratio (0.0615) combined with thick, non-uniform walls (1.36–46.75mm), 52 undercuts, and a mix of cylindrical bores/bosses (bearing seat, ball-joint and tie-rod bores) is fully consistent with a near-net sand-cast ductile-iron part that is subsequently CNC-machined on 3 setups to finish bores, mounting pads and threaded holes. Given the load-bearing, drivetrain-adjacent nature of the part and no aluminium/thin-wall HPDC signal, ductile iron (GJS500/600) is recommended over aluminium, in line with real-world steering knuckle production practice. At 200,000 units/year, high-throughput green-sand or shell-moulding lines with a durable metal pattern (£17k) are economically favoured over die costs for HPDC/forging.

§9 — Analysis Limitations & Assumptions

1. Material (ductile iron vs steel forging vs aluminium) is inferred from part type and geometry signals, not from explicit metadata — treat as medium confidence.
2. Cavity count and pattern life are rule-of-thumb estimates for a high-volume automotive sand-casting line, not derived from OCCT geometry.
3. Cost range assumes UK labour rates and standard sand-casting + 3-axis/5-axis finish machining; actual costs will vary with supplier tooling amortisation strategy.

Stage-1 pre-selection: cast_and_machine (100%) ·